

AMAZON SALES DATA ANALYSIS USING SQL

Prepared By

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Role

Aspiring Data Analyst

Project Type

SQL Data Analysis Project

Executive Summary

The objective of this project was to analyze Amazon sales data using SQL and generate meaningful business insights from product ratings, pricing, discounts, customer reviews, and category performance.

The dataset was imported from an Excel spreadsheet into Microsoft SQL Server Management Studio (SSMS). A total of 17 SQL queries were executed to perform data exploration, filtering, aggregation, and analytical reporting.

This project demonstrates practical SQL skills commonly used in Data Analyst roles.

1. Introduction

E-commerce platforms generate large volumes of product-related data that can be analyzed to understand customer preferences, product performance, and pricing strategies.

This project focuses on analyzing Amazon product data using SQL. The analysis includes customer ratings, review counts, discount structures, category performance, and product pricing.

The project aims to transform raw data into actionable business insights.

2. Project Objectives

The primary objectives of this project are:

- Explore the Amazon sales dataset.
 - Analyze customer ratings and reviews.
 - Identify highly rated products.
 - Evaluate pricing and discount strategies.
 - Measure customer engagement through review analysis.
 - Perform category-wise product analysis.
 - Detect missing or invalid data.
 - Demonstrate SQL-based analytical skills.
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3. Dataset Overview

Dataset Source

Amazon Sales Dataset (Excel Spreadsheet)

Dataset Size

- Total Records: 1465
- Total Columns: 16

Important Columns

Column Name	Description
Product Name	Product title
Category	Product category
Actual Price	Original product price
Discounted Price	Selling price after discount
Discount Percentage	Percentage discount offered
Rating	Customer rating
Rating Count	Number of customer reviews

4. Data Preparation

Before analysis, the dataset was imported into Microsoft SQL Server.

Several data quality checks were performed:

- Numeric conversions using TRY_CAST().

- Identification of invalid rating values.
- Cleaning review count values using REPLACE().
- Validation of price and rating fields.

These preprocessing steps improved analysis accuracy.

5. SQL Analysis

Query 1 – Display Entire Dataset

Purpose : To examine all available records and verify successful data import.

SQL Query:

```
select * from amazon;
```

Result:

product_id	product_name	category	discounted_price	actual_price	discount_percentage	rating	rating_count	about_product
B07JW9H4J1	Wayona Nylon Braided USB to Lightning Fast Charging ...	Computers&Accessories Accessories&Peripherals Cable...	399	1099	0.64	4.2	24269	High Compatibility : C
B098NS6PVG	Ambrane Unbreakable 60W / 3A Fast Charging 1.5m Br...	Computers&Accessories Accessories&Peripherals Cable...	199	349	0.43	4	43994	Compatible with all T
B096MSW6CT	Source Fast Phone Charging Cable & Data Sync USB ...	Computers&Accessories Accessories&Peripherals Cable...	199	1899	0.9	3.9	7928	【 Fast Charger& Data
B08HJ86NZ	boAt Deuce USB 300 2 in 1 Type-C & Micro USB Stres...	Computers&Accessories Accessories&Peripherals Cable...	329	699	0.53	4.2	94363	The boAt Deuce US
B08CF3B7N1	Potronics Konnect L 1.2M Fast Charging 3A 8 Pin USB...	Computers&Accessories Accessories&Peripherals Cable...	154	399	0.61	4.2	16905	[CHARGE & SYNC F
B08Y1TFS66	pTron Solero TB301 3A Type-C Data and Fast Chargin...	Computers&Accessories Accessories&Peripherals Cable...	149	1000	0.85	3.9	24871	Fast Charging & Data
B08WRWPM22	boAt Micro USB 55 Tangle-free, Sturdy Micro USB Cabl...	Computers&Accessories Accessories&Peripherals Cable...	176.63	499	0.65	4.1	15188	It Ensures High Spee
B08DDRGWTJ	MI Usb Type-C Cable Smartphone (Black)	Computers&Accessories Accessories&Peripherals Cable...	229	299	0.23	4.3	30411	1m long Type-C USB
B008IFXQFU	TP-Link USB WiFi Adapter for PC(TL-WN725N), N150 ...	Computers&Accessories NetworkingDevices Network A...	499	999	0.5	4.2	179691	USB WiFi Adapter

Observation : The dataset contains product, pricing, discount, rating, and review information.

Query 2 – Product Name and Rating

Purpose : To display customer ratings for each product.

SQL Query :

```
SELECT product_name, rating
```

```
FROM amazon;
```

Results Messages

	product_name	rating
1	Wayona Nylon Braided USB to Lightning Fast Charging ...	4.2
2	Ambrane Unbreakable 60W / 3A Fast Charging 1.5m Br...	4
3	Sounce Fast Phone Charging Cable & Data Sync USB ...	3.9
4	boAt Deuce USB 300 2 in 1 Type-C & Micro USB Stress...	4.2
5	Portronics Konnect L 1.2M Fast Charging 3A 8 Pin USB...	4.2
6	pTron Solero TB301 3A Type-C Data and Fast Charging...	3.9
7	boAt Micro USB 55 Tangle-free, Sturdy Micro USB Cabl...	4.1
8	MI Usb Type-C Cable Smartphone (Black)	4.3
9	TP-Link USB WiFi Adapter for PC(TL-WN725N), N150 ...	4.2
10	Ambrane Unbreakable 60W / 3A Fast Charging 1.5m Br...	4

Observation : Ratings help evaluate customer satisfaction levels.

Query 3 – Products with Rating Above 4

Purpose : To identify highly rated products.

SQL Query:

```
SELECT * FROM amazon
WHERE TRY_CAST(rating AS FLOAT) > 4;
```

Result:

product_id	product_name	category	discounted_price	actual_price	discount_percentage	rating	rating_count	about_product
B07JW9H4J1	Wayona Nylon Braided USB to Lightning Fast Charging ...	Computers&Accessories/Accessories&Peripherals/Cable...	399	1099	0.64	4.2	24269	High Compatibility : C
B08HJD96NZ	boAt Deuce USB 300 2 in 1 Type-C & Micro USB Stres...	Computers&Accessories/Accessories&Peripherals/Cable...	329	699	0.53	4.2	94363	The boAt Deuce US
B08CF3B7N1	Portronics Konnect L 1.2M Fast Charging 3A 8 Pin USB...	Computers&Accessories/Accessories&Peripherals/Cable...	154	399	0.61	4.2	16905	[CHARGE & SYNC F
B08WRWPM22	boAt Micro USB 55 Tangle-free, Sturdy Micro USB Cabl...	Computers&Accessories/Accessories&Peripherals/Cable...	176.63	499	0.65	4.1	15188	It Ensures High Spee
B08DDRGWTJ	MI Usb Type-C Cable Smartphone (Black)	Computers&Accessories/Accessories&Peripherals/Cable...	229	299	0.23	4.3	30411	1m long Type-C USB
B088FXQFU	TP-Link USB WiFi Adapter for PC(TL-WN725N), N150 ...	Computers&Accessories/NetworkingDevices/NetworkA...	499	999	0.5	4.2	179691	USB WiFi Adapter —
B07VHS7MZ1	iQOO 9 SE 5G (Sunset Sierra, 8GB RAM, 128GB Stora...	Electronics/Mobiles&Accessories/Smartphones&BasicM...	29990	39990	0.25	4.3	8399	Qualcomm Snapdrag
B08MM7R9ZG	Noise Colorfit Smart Watch Charger 2 Pin USB Fast Ch...	Electronics/Wearable Technology/Smart Watches	249	999	0.75	4.5	38	Specially designed fo
B08M66K48D	POPIO Tempered Glass Screen Protector Compatible fo...	Electronics/Mobiles&Accessories/MobileAccessories/Ma...	299	599	0.5	4.3	4674	Hardness: High hard

Observation: Products rated above 4 are generally considered well-received by customers.

Query 4 – Products with Discounted Price Below ₹500

Purpose: To identify affordable products.

SQL Query :

```
SELECT product_name, discounted_price
FROM amazon
```

```
WHERE TRY_CAST(discounted_price AS DECIMAL(10,2)) < 500;
```

Result:

	product_name	discounted_price
1	Wayona Nylon Braided USB to Lightning Fast Charging ...	399
2	Ambrane Unbreakable 60W / 3A Fast Charging 1.5m Br...	199
3	Sounce Fast Phone Charging Cable & Data Sync USB ...	199
4	boAt Deuce USB 300 2 in 1 Type-C & Micro USB Stress...	329
5	Portronics Konnect L 1.2M Fast Charging 3A 8 Pin USB...	154
6	pTron Solero TB301 3A Type-C Data and Fast Charging...	149
7	boAt Micro USB 55 Tangle-free, Sturdy Micro USB Cabl...	176.63
8	MI Usb Type-C Cable Smartphone (Black)	229
9	TP-Link USB WiFi Adapter for PC(TL-WN725N), N150 ...	499
10	Ambrane Unbreakable 60W / 3A Fast Charging 1.5m Br...	199

Observation: These products may attract price-sensitive customers.

Query 5 – Top 10 Highest Rated Products

Purpose: To identify top-performing products.

SQL Query :

```
SELECT TOP 10 product_name, rating
FROM amazon
ORDER BY TRY_CAST(rating AS FLOAT) DESC;
```

Result:

	product_name	rating
1	Amazon Basics Wireless Mouse 2.4 GHz Connection,...	5
2	Syncwire LTG to USB Cable for Fast Charging Compati...	5
3	REDTECH USB-C to Lightning Cable 3.3FT, [Apple M...	5
4	Swiffer Instant Electric Water Heater Faucet Tap Hom...	4.8
5	Oratech Coffee Frother electric, milk frother electric, co...	4.8
6	Instant Pot Air Fryer, Vortex 2QT, Touch Control Panel,...	4.8
7	FIGMENT Handheld Milk Frother Rechargeable, 3-Spe...	4.7
8	Zuvexa USB Rechargeable Electric Foam Maker - Ha...	4.7
9	Multifunctional 2 in 1 Electric Egg Boiling Steamer Egg ...	4.7
10	Campfire Spring Chef Prolix Instant Portable Water Hea...	4.7

Observation: These products demonstrate the highest customer satisfaction.

Query 6 – Maximum Actual Price

Purpose: To find the most expensive product in the dataset.

SQL Query

```
SELECT MAX(actual_price) AS MaxPrice
FROM amazon;
```

Result:

Results		Messages	
	MaxPrice		
1	9999		

Observation : Represents premium-priced products.

Query 7 – Minimum Discounted Price

Purpose: To identify the least expensive product after discount.

SQL Query :

```
SELECT MIN(discounted_price) AS MinPrice
FROM amazon;
```

Result:

Results		Messages	
	MinPrice		
1	100		

Observation: Represents entry-level pricing opportunities.

Query 8 – Average Product Rating

Purpose: To calculate overall customer satisfaction.

SQL Query :

```
SELECT AVG(TRY_CAST(rating AS FLOAT)) AS AvgRating
FROM amazon;
```

Result:

Results		Messages	
	AvgRating		
1	4.09658469945355		

Observation: Provides a general measure of product quality perception.

Query 9 – Total Number of Products

Purpose : To determine dataset size.

SQL Query :

```
SELECT COUNT(*) AS TotalProducts
FROM amazon;
```

Result:

Results		Messages	
	TotalProducts		
1	1465		

Observation : The dataset contains 1465 product records.

Query 10 – Unique Products

Purpose : To identify distinct product names.

SQL Query :

```
SELECT DISTINCT product_name
FROM amazon;
```

Result :

	product_name
1	!!1000 Watt/2000-Watt Room Heater!! Fan Heater!!Pur...
2	!!HANEUL!!1000 Watt/2000-Watt Room Heater!! Fan ...
3	10k 8k 4k HDMI Cable, Certified 48Gbps 1ms Ultra Hig...
4	10WeRun Id-116 Bluetooth Smartwatch Wireless Fitnes...
5	3M Post-it Sticky Note Cube, 200 Sheets (4 Colors x 50 ...
6	3M Scotch Double Sided Heavy Duty Tape(1m holds 4....
7	4 in 1 Handheld Electric Vegetable Cutter Set,Wireless ...
8	7SEVEN Compatible LG TV Remote Suitable for LG No...
9	7SEVEN® Bluetooth Voice Command Remote for Xiao...
10	7SEVEN® Compatible for Mi tv Remote Control Original ...

Observation : Useful for removing duplicates during analysis.

Query 11 – Category-wise Product Count

Purpose : To analyse product distribution across categories.

SQL Query :

```
SELECT category, COUNT(*) AS TotalProducts
FROM amazon
GROUP BY category
ORDER BY TotalProducts DESC;
```

Result:

	category	TotalProducts
1	Computers&Accessories Accessories&Peripherals Cables...	233
2	Electronics Wearable Technology Smart Watches	76
3	Electronics Mobiles&Accessories Smartphones&BasicMo...	68
4	Electronics Home Theater,TV&Video Televisions Smart Te...	63
5	Electronics Headphones,Earbuds&Accessories Headpho...	52
6	Electronics Home Theater,TV&Video Accessories Remot...	49
7	Home&Kitchen Kitchen&Home Appliances Small KitchenA...	27
8	Home&Kitchen Kitchen&Home Appliances Vacuum,Clean...	24
9	Electronics Home Theater,TV&Video Accessories Cables ...	24
10	Computers&Accessories Accessories&Peripherals Keybo...	24

Query executed successfully.

Observation : Some categories contain significantly more products than others.

Query 12 – Top 10 Most Reviewed Products

Purpose : To identify products receiving maximum customer engagement.

SQL Query :

```
SELECT TOP 10 product_name, rating_count
FROM amazon
ORDER BY rating_count DESC;
```

Result:

Results Messages		
	product_name	rating_count
1	IT2M Designer Mouse Pad for Laptop/Computer (9.2 ...	9998
2	AirCase Protective Laptop Bag Sleeve fits Upto 13.3" ...	9940
3	Eureka Forbes car Vac 100 Watts Powerful Suction V...	992
4	Eveready Red 1012 AAA Batteries - Pack of 10	989
5	Sennheiser CX 80S in-Ear Wired Headphones with in-l...	98250
6	Wecool Nylon Braided Multifunction Fast Charging Ca...	9792
7	Wecool Nylon Braided Multifunction Fast Charging Ca...	9792
8	Pigeon By Stovekraft ABS Plastic Acer Plus Induction ...	9791
9	Bajaj DHX-9 1000W Heavy Weight Dry Iron with Adv...	9772
10	Pigeon 1.5 litre Hot Kettle and Stainless Steel Water B...	976

✓ Query executed successfully.

Observation : High review counts generally indicate strong market interest.

Query 13 – Products with More Than 50% Discount

Purpose : To identify heavily discounted products.

SQL Query :

```
SELECT product_name, discount_percentage
FROM amazon
WHERE discount_percentage > 0.50;
```

Result:

Results		Messages
	product_name	discount_percentage
1	Wayona Nylon Braided USB to Lightning Fast Charging ...	0.64
2	Source Fast Phone Charging Cable & Data Sync USB ...	0.9
3	boAt Deuce USB 300 2 in 1 Type-C & Micro USB Stress...	0.53
4	Portronics Konnect L 1.2M Fast Charging 3A 8 Pin USB...	0.61
5	pTron Solero TB301 3A Type-C Data and Fast Charging...	0.85
6	boAt Micro USB 55 Tangle-free, Sturdy Micro USB Cabl...	0.65
7	SHREENOVA ID116 Plus Bluetooth Fitness Smart Wat...	0.86
8	Noise_Colorfit Smart Watch Charger 2 Pin USB Fast Ch...	0.75
9	10WeRun Id-116 Bluetooth Smartwatch Wireless Fitnes...	0.74
10	Tokdis MX-1 Pro Bluetooth Calling Smartwatch - 1.69" L...	0.74

Query executed successfully.

Observation : These products may be part of promotional campaigns.

Query 14 – Search Products Containing 'USB'

Purpose : To perform keyword-based product analysis.

SQL Query :

```
SELECT * FROM amazon
```

```
WHERE product_name LIKE '%USB%';
```

Result:

Results		Messages							
	product_id	product_name	category	discounted_price	actual_price	discount_percentage	rating	rating_count	about_product
1	B07JW9H4J1	Wayona Nylon Braided USB to Lightning Fast Charging ...	Computers&Accessories/Accessories&Peripherals/Cable...	399	1099	0.64	4.2	24269	High Compatibility : C
2	B096MSW6CT	Source Fast Phone Charging Cable & Data Sync USB ...	Computers&Accessories/Accessories&Peripherals/Cable...	199	1899	0.9	3.9	7928	Fast Charger& Data
3	B08HJ86NZ	boAt Deuce USB 300 2 in 1 Type-C & Micro USB Stres...	Computers&Accessories/Accessories&Peripherals/Cable...	329	699	0.53	4.2	94363	The boAt Deuce US
4	B08CF3B7N1	Portronics Konnect L 1.2M Fast Charging 3A 8 Pin USB...	Computers&Accessories/Accessories&Peripherals/Cable...	154	399	0.61	4.2	16905	[CHARGE & SYNC F
5	B08Y1TFSP6	pTron Solero TB301 3A Type-C Data and Fast Chargin...	Computers&Accessories/Accessories&Peripherals/Cable...	149	1000	0.85	3.9	24871	Fast Charging & Data
6	B08WRWPM22	boAt Micro USB 55 Tangle-free, Sturdy Micro USB Cabl...	Computers&Accessories/Accessories&Peripherals/Cable...	176.63	499	0.65	4.1	15188	It Ensures High Spec
7	B08DDRGWTJ	Mi Usb Type-C Cable Smartphone (Black)	Computers&Accessories/Accessories&Peripherals/Cable...	229	299	0.23	4.3	30411	1m long Type-C USB
8	B008IFXQFU	TP-Link USB WiFi Adapter for PC(TL-WN725N), N150 ...	Computers&Accessories/NetworkingDevices/NetworkA...	499	999	0.5	4.2	179691	USB WiFi Adapter —
9	B082LZGK39	Ambrane Unbreakable 60W / 3A Fast Charging 1.5m Br...	Computers&Accessories/Accessories&Peripherals/Cable...	199	299	0.33	4	43994	Universal Compatibil

Query executed successfully.

DESKTOP-C1488JH\SPARTA (16...)|DESKTOP-C1488JH\HP (65)|AmazonProject|00:00:00|328 rows

Observation : Useful for filtering specific product types.

Query 15 – Invalid or NULL Ratings

Purpose : To identify data quality issues.

SQL Query :

```
SELECT * FROM amazon
```

WHERE TRY_CAST(rating AS FLOAT) IS NULL;

Result:

	product_id	product_name	category	discounted_price	actual_price	discount_percentage	rating	rating_count	about_product
1	B08L12N5H1	Eureka Forbes car Vac 100 Watts Powerful Suction...	Home&Kitchen/Kitchen&HomeAppliances/Vacuum,Clean...	2099	2499	0.16	1	992	No Installation is provided for

Query executed successfully. DESKTOP-C1488JH-SPARTA (16... | DESKTOP-C1488JH-VHP (65) | AmazonProject | 00:00:00 | 1 rows

Observation : Missing values should be addressed before advanced analytics.

Query 16 – Savings Analysis

Purpose : To calculate customer savings.

SQL Query :

```
SELECT product_name,
       actual_price,
       discounted_price,
       TRY_CAST(actual_price AS DECIMAL(10,2))
       - TRY_CAST(discounted_price AS DECIMAL(10,2)) AS Savings
FROM amazon;
```

Result:

	product_name	actual_price	discounted_price	Savings
1	Wayona Nylon Braided USB to Lightning Fast Charging ...	1099	399	700.00
2	Ambrane Unbreakable 60W / 3A Fast Charging 1.5m Br...	349	199	150.00
3	Sounce Fast Phone Charging Cable & Data Sync USB ...	1899	199	1700.00
4	boAt Deuce USB 300 2 in 1 Type-C & Micro USB Stress...	699	329	370.00
5	Portronics Konnect L 1.2M Fast Charging 3A 8 Pin USB...	399	154	245.00
6	pTron Solero TB301 3A Type-C Data and Fast Charging...	1000	149	851.00
7	boAt Micro USB 55 Tangle-free, Sturdy Micro USB Cabl...	499	176.63	322.37
8	MI Usb Type-C Cable Smartphone (Black)	299	229	70.00
9	TP-Link USB WiFi Adapter for PC(TL-WN725N), N150 ...	999	499	500.00
10	Ambrane Unbreakable 60W / 3A Fast Charging 1.5m Br...	299	199	100.00

Query executed successfully.

Observation : The difference between actual and discounted prices represents customer benefit.

Query 17 – Category-wise Review Analysis

Purpose : To measure customer engagement across categories.

SQL Query :

```
SELECT category,
       SUM(
           TRY_CAST(REPLACE(rating_count, ',', '') AS BIGINT)
       ) AS TotalReviews
FROM amazon
GROUP BY category

ORDER BY TotalReviews DESC;
```

Result:

	category	TotalReviews
1	Electronics Headphones,Earbuds&Accessories Headph...	4204939
2	Computers&Accessories Accessories&Peripherals Cable...	3547816
3	Electronics Mobiles&Accessories Smartphones&BasicM...	2493269
4	Electronics Home Theater,TV&Video Accessories Cables...	1906054
5	Electronics Wearable Technology Smart Watches	1644476
6	Electronics Accessories MemoryCards Micro SD	1113592
7	Computers&Accessories ExternalDevices&DataStorage ...	774744
8	Electronics Home Theater,TV&Video Televisions Smart T...	760279
9	Computers&Accessories NetworkingDevices NetworkA...	727113
10	Electronics Mobiles&Accessories MobileAccessories Ch...	688810

✓ Query executed successfully.

Observation : Categories with higher review counts indicate stronger customer interaction.

6. Key Business Insights

The analysis generated the following insights:

- Highly rated products indicate strong customer satisfaction.
 - Large discounts increase product attractiveness.
 - Some categories dominate the marketplace in terms of product volume.
 - Review counts reveal customer engagement patterns.
 - Data quality validation is essential before performing advanced analytics.
 - Savings analysis highlights the effectiveness of discount strategies.
-

7. SQL Skills Demonstrated

- SELECT Statements
 - WHERE Clause Filtering
 - ORDER BY Sorting
 - Aggregate Functions
 - COUNT()
 - AVG()
 - MAX()
 - MIN()
 - SUM()
 - DISTINCT
 - GROUP BY
 - LIKE Operator
 - TRY_CAST()
 - REPLACE()
 - Data Cleaning
 - Data Validation
-

8. Conclusion

This project successfully analyzed Amazon sales data using SQL and generated valuable insights related to customer ratings, discounts, pricing, product popularity, and category performance.

The project demonstrates practical SQL skills required for Data Analyst positions and provides experience in data exploration, cleaning, aggregation, and business reporting.

9. Future Scope

Future improvements may include:

- Power BI Dashboard Development
- Interactive Visualizations
- Advanced SQL Reporting
- Customer Segmentation
- Product Trend Analysis
- Predictive Analytics
- Sales Forecasting